

R scripts and Data for Spline Analysis of Whole Lake Nutrient Additions to Peter Lake September 25, 2026

R scripts and data to repeat the analysis for the paper “Quantifying Phytoplankton States and Thresholds from Whole Lake Nutrient Additions” by Michael L. Pace, Stephen R. Carpenter, and Grace M. Wilkinson are available in the GitHub repository cited in the main text.

Spline Analysis

The spline analysis uses four scripts run sequentially. Each sequence creates an analysis for one year for one of four variables (dissolved oxygen saturation, chlorophyll-a (logged), phycocyanin (referred to as BGA, also logged), and pH. The example sequence below is for the variable dissolved oxygen saturation in Peter Lake, year 2015. Data for the other variables used in the study are provided.

The sequence of steps is:

1) Calculate daily values from high frequency data using the first script:

DailyDataStats_DOsat.R

This program provides medians, 5th percentiles, 95th percentiles, and range for every day of the time series for dissolved oxygen percent saturation. The program will need to be modified to make similar calculations for pH or phycocyanin (referred to as BGA).

Comments in the script provide guidance on selections needed to run the program. For analysis of chlorophyll this script can be skipped as data files are provided for each year.

Daily chlorophyll was directly measured and provided as:

Daily_R_Ch12013.Rdata

Daily_R_Ch12014.Rdata

Daily_R_Ch12015.Rdata

Daily_R_Ch12019.Rdata

Daily_R_Ch12024.Rdata

Input files are high frequency data referred to as *Project1_HYLB.Rdata*, *Project2_YSI.Rdata*, *Project2_HYLB*, *Project3_YSI_Paul.Rdata*, *Project3_YSI_Peter.Rdata*. Project 1 is data for 2013-2015; Project 2 is data for 2019; Project 3 is data for 2024. Comments in the program provide guidance on how to select data for the lake, year, and variable of interest.

Outputs:

File name (Fname) is: *Daily_R_DOsat2015.Rdata* for the example provide.

2) Load the output file from step 1 in the next script:

Langevin1.DLM.DOsat

This program runs a dynamic linear model and other diagnostics. The program calls *ODLMAR_NoBoot_2018-10-20.R* for the online estimation of AR(p) – see comments in program.

Outputs:

File name (Fname) is: *DLM_DOsat_medium_Peter15.Rdata* for the example provide.

3) Load the output file from step2 in the next script:

Langevin2_ADF_stat_Markov_DOsat.R

This program analyzes autocorrelation and applies an augmented Dickey-Fuller test. There is no output file. Key results are printed.

4) Load an input file (example: *DLM_DOsat_median_Peter15.Rdata*) into the next script:

Langevin3_B-spline_DOsat.R

This program fits a B-spline to give deterministic equilibria and calculates the effective potential to give stochastic equilibria (see explanation in paper). Choices for knots, splines, and data trimming are made by the analyst – see text for guidance – and Table S2. The program calls *EPFunction+EQ_ssML_nKnot.R* to estimate the effective potential curve.

Outputs:

File name (Fname) is: *Bspline_DOsat_median_Peter15.Rdata*

Steps 1-4 provide the core results for the paper. The data and scripts used to generate the figures are summarized below in the order they appear.

Bootstrap Estimates of Standard Deviations of Equilibria

Uncertainty of the equilibria are estimated by bootstrapping the estimates of the spline model. The bootstrap calculation is added to the original B-spline program and has the name: *DOsat_Bspline+smoothspline_Langevin3+Boot.R*

The choices of knots, splines and data trimming are the same as used in the original analysis. The number of bootstrap cycles is set by ‘nboot’ in the program. Because full bootstrapping runs require a few hours on a standard computer, a good practice is to test with nboot=10. Final bootstrap results are based on nboot=1000

The input file using the Peter 2015 example is *DLM_DOsat_median_Peter15.Rdata*

Output file is

Boot_1000_Spline_DOsat_Peter2015

To analyze bootstrap results load the output file in *After-Boot_DOsat_stats_densities_Peter_2015_Example_Supplement.R*

This program produces plots of the frequencies of equilibria, D1 D2 and effective potential curves for all bootstrap cases, means and standard deviations of equilibria for cases with 3 equilibria, density plots, and bias estimates.

Figures – Input Data and Figure files

Figure 1 is generated from two output files of the daily data – step 1 above. These files are:

Daily_R_DOsat2015.Rdata

Daily_R_ChI2015.Rdata

The script is:

Figure 1 2015 DO and ChI

Figure 2 is generated from results of the B-spline program and a script for making Figure 2

Figure 2 DOsat_Bspline+smoothspline_Langevin3

Figure 3 is generated using four output files of the daily data – step 1 above. These files are:

Daily_R_DOsat2013.Rdata

Daily_R_DOsat2014.Rdata

Daily_R_DOsat2019.Rdata

Daily_R_DOsat2024.Rdata

The script is:

Figure 3 DOsat Plots of Daily Medians

Figure 4 is generated from saved files of the effective potential results from the B-spline program. These files are:

EP_R13.Rdata

EP_R14.Rdata

EP_R15.Rdata

EP_R19.Rdata

EP_R24.Rdata

The script is:

Figure 4 DOsat EFP Plots

Figure 5 is generated from saved files of equilibria using the B-spline program and mean equilibria and standard deviations from the *After-Boot* program (see above). The input files are:

DOsat_Equilibria.csv

DOsat_EQ_Mean+SD.csv

The script is:

Figure 5 Equilibria Data + Bootstrap.R

Figure 6 is generated from saved files of the effective potential results from the B-spline program. These files are:

EP_R15_LogChl.Rdata

EP_R19_LogChl.Rdata

EP_R24_LogChl.Rdata

EP_R15_LogBGA.Rdata

EP_R19_LogBGA.Rdata

EP_R24_LogBGA.Rdata

EP_R15_pH.Rdata

EP_R19_pH.Rdata

EP_R24_pH.Rdata

The script is:

Figure 6 EFP plots for Chl BGA pH.R

Figure 7 is generated by compiling results on the day of year crossing a threshold determined in *Langevin3_Bspline_DOsat.R*. The threshold data are in the script.

Daily data as CSV files for the years 2015, 2019, or 2004 are input as time series in *Langevin3_Bspline_DOsat.R* for comparing with the unstable threshold. Example is *Daily_R_DOsat2015.csv*.

The Figure 7 script is Figure 7 Bar Plots DOY

Tables

Table 1 on nutrient additions compiled information from prior publications: 2013-2015 additions - Wilkinson et al. 2018; 2019 addition - Buelo et al. 2022; 2024 addition previously unpublished information.

Supplementary Information

The description below is ordered by the subheadings of Supplement

Lake physical and chemical conditions – Means for the lakes and years can be generated from:

Wilkinson, G., S. Carpenter, and M. Pace. 2026. Cascade Project at North Temperate Lakes LTER Core Data Carbon 1984 - 2025 ver 9. Environmental Data Initiative. <https://doi.org/10.6073/pasta/ae1c3e5ed5f47836576a5d58a7b9d83c> (Accessed 2026-09-30).

Wilkinson, G., S. Carpenter, and M. Pace. 2026. Cascade Project at North Temperate Lakes LTER Core Data Nutrients 1991 - 2025 ver 6. Environmental Data Initiative. <https://doi.org/10.6073/pasta/5ef427415a5b54b27cb9b1b6edc71a92> (Accessed 2026-09-30).

Wilkinson, G., S. Carpenter, and M. Pace. 2026. Cascade Project at North Temperate Lakes LTER Core Data Process Data 1984 - 2025 ver 6. Environmental Data Initiative. <https://doi.org/10.6073/pasta/547a1afe2995f9473db8af6f6c16caa2> (Accessed 2026-09-30).

B-spline parameters and data treatment – Table S2 is from the Langevin3_Bspline program as run for individual variables and years.

Bootstraps estimates of uncertainty – Figures S1 and S2 are generated in the program *After-Boot_DOsat_stats_densities_Peter_2015_Example_Supplement.R*

Simulation Test of Spline Method – The programs in Brock et al. (2026) reproduce the results along with the model from the Langevin3-Bspline program.

Paul Lake Effective Potential – The four programs described under the “Spline Analysis” heading above generate the results for Paul Lake. The main modification is to calculate the daily values for Paul Lake by year and use these data in subsequent steps.

Uncertainty in Date of Threshold Crossing – The After-Boot program was used to generate the pseudo-data on equilibria to estimate mean, standard deviation, and coefficient of variation for “Pseudo-Crossing (DOYs)”.

References

Brock, WA, S.R. Carpenter SR, Pace ML, Wilkinson GM. 2026. Ecosystem Experiments: How does manipulation rate affect indicators of resilience? *Ecosystems* 29: 23

Buelo, CD, Pace ML, Carpenter SR, Stanley EH, Ortiz DA, Ha DT. 2022. Evaluating the performance of temporal and spatial early warning statistics of algal blooms. *Ecological Application* 32:e2616

Wilkinson GM, Carpenter SR, Cole JJ, Pace ML, Batt RD, Buelo CD, Kurtzweil JT. 2018. Early warnings signals precede cyanobacterial blooms in multiple whole-lake experiments. *Ecological Monographs* 88: 188-203